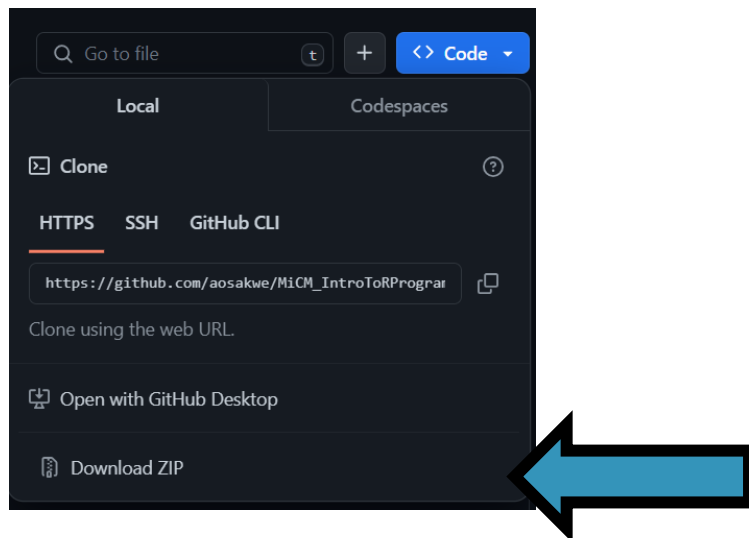
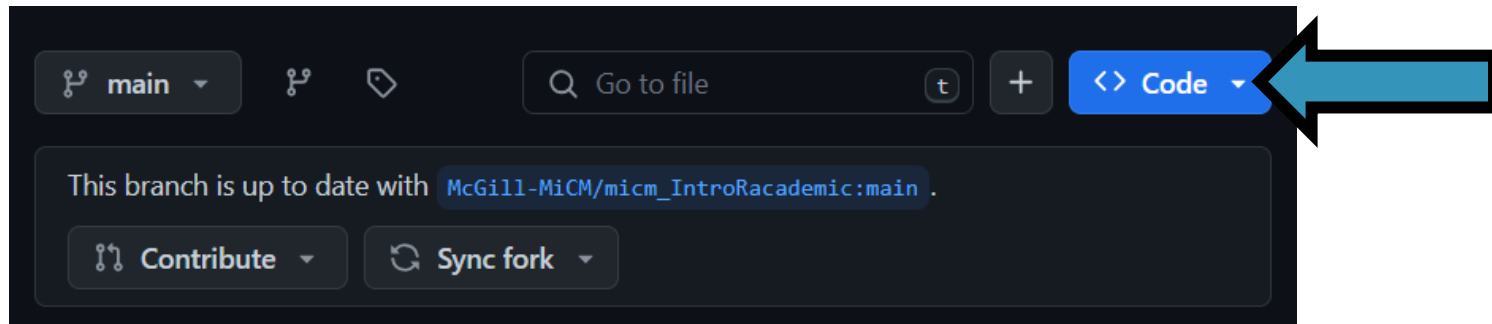


Download Workshop Materials

1. Go to <https://github.com/QLS-MiCM/IntroToR>

2.



Intro to programming in R

Lead: Thomas Zheng
Facilitator: Leighton Smith

Feb 9, 2026

Materials adapted from A. Osakwe, L. Soto, X. Xie, and M. Yang.

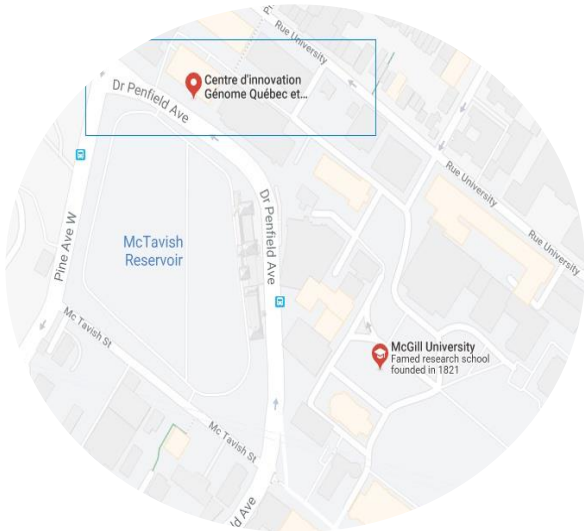


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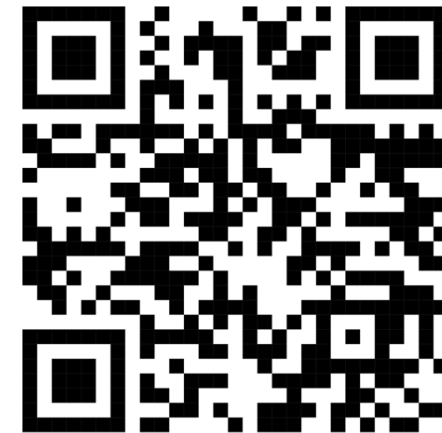
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Mission statement: deliver quality workshops designed to help biomedical researchers develop the skills they need to succeed.



Location: 550 Sherbrooke West,
Montreal, Quebec



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Workshop	Date	Location	Registration
How to Think in Code	September 22, 1 - 3 PM	Sherbrooke 550 W, Rm 136	Closed
Intro to Git & GitHub	September 23, 1 - 3 PM	Sherbrooke 550 W, Rm 136	Closed
Intro to Unix	September 25, 1 - 3 PM	Sherbrooke 550 W, Rm 136	Closed
Intro to DRAC/Compute Canada	September 28, 10 AM - 12 PM	Education Building, 3700 McTavish, Rm 133	Open
Intro to R	September 30, 1 - 3 PM	Sherbrooke 550 W, Rm 136	Open
Intro to Python (Part 1)	October 2, 1 - 3 PM	Sherbrooke 550 W, Rm 189	Open
More workshops to be announced soon!			

<https://www.mcgill.ca/micm/training/workshops-series>



Workshop outline Part 1

1 The language

History
Foundation
Syntax
Logical ops
Help
Packages

2 Data types

Vectors
Factors
Lists
Data Frames
Arrays
Hands on

3 Control Structures

Functions
If statement
for loop
Hands on



Workshop outline Part 2

4

**Basic data
manipulation**

Read & Write
Subset
Split
Join

Hands on

5

**Advanced data
Manipulation**

dplyr
tidyr
plyr
DataTable

Hands on

6

**Generating
Outputs**

Graphics
ggplot2
RMarkdown
Templates

7

**Software
development**

Good coding
practices
Documentation
standards
Debugging



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Workshop Components

- Theory
- Code Examples
- Hands-on Activities



1. The R programming language

Learning objectives

- Why Excel is not enough
- What is R
- What is an IDE
- Basic Operations



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Why not Excel?



- Easy at first glance
- **Issues**
 1. Hard to automate
 2. Hard to reproduce
 3. Inflexible
 4. Slow!



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- **Statistical** Programming Language
- Integrated suite for **data manipulation, analysis, and graphical visualization**
- Environment where **statistical tests can be performed**
- Its functionality can be easily extended with ***packages***
- GNU project of free software
- Users have the freedom to:
 - Run the program
 - View and modify the source code
 - Redistribute copies and
 - Distribute their modifications

R facts

- Interpreted language
- Object-oriented
- No spaces allowed in variable names
- Case sensitive
- 1-based indexing
- Allows user-defined functions
- Works with environments



R Files

- Many types of files can contain R code
 - **.R ‘Script’**
 - **.Rmd ‘R Notebook’**
 - .qmd ‘Quarto Notebook’
 - .ipynb ‘Jupyter Notebook’
- Scripts
 - Automation & Portability
- Notebooks
 - Documentation
 - Accessibility



R & RStudio



- R & RStudio are **different entities**
- R is the programming language
 - The actual code we execute
 - Developed at the University of Auckland
- RStudio is an **Integrated Development Environment (IDE)**
 - A GUI software to develop and execute R code
 - Developed by Posit



```
PS C:\Program Files\R\R-4.2.2\bin> .\R.exe
```

```
R version 4.2.2 (2022-10-31 ucrt) -- "Innocent and Trusting"  
Copyright (C) 2022 The R Foundation for Statistical Computing  
Platform: x86_64-w64-mingw32/x64 (64-bit)
```

```
R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.
```

```
  Natural language support but running in an English locale
```

```
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.
```

```
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.
```

```
> print('hello world')  
[1] "hello world"  
> |
```



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Source Pane

Environment Pane

The screenshot shows the RStudio interface with four panes highlighted by colored borders and arrows:

- Source Pane (Red border):** Displays the R Markdown source file. It includes a header with affiliation, date, and output format settings. Below is a code chunk for the McGill initiative logo, which has been rendered as an image. The title "Introduction to R" and a paragraph of introductory text are also visible.
- Environment Pane (Pink border):** Shows the current environment with two variables: `var1` with value `"mango"` and `var2` with value `"mangos"`.
- Console Pane (Yellow border):** Shows the R console output. It displays the execution of several R commands, including creating vectors, performing arithmetic and logical operations, and assigning values to variables. The last command executed is `var1 < var2`, which returned `TRUE`.
- Files/Plot/Help Pane (Purple border):** Shows the file explorer for the current project. It lists various files and folders, including `.Rhistory`, `data`, `Install_R_Rstudio.pdf`, `MICM_IntroToR.pdf`, `MICM_IntroToRProgramming.Rproj`, `MICM_outline_IntroR.docx`, `README.rmd`, `scripts`, and `MICM_IntroToR.pptx`.

Console Pane

Files/Plot/Help Pane



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Data types and data structures

Learning objectives

- Understand the differences between classes, objects and data types in R
- Create objects of different types
- Subset and index objects
- Learn and use vectorized operations



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Atomic Classes

Also called data types

Character	A,b,c,d,e,..
Numeric (real numbers)	1.00,2.00,... Inf, NaN
Integer	1L,2L,3L,4L,....
Complex	2i
Logical (True/False)	TRUE,FALSE
Missing Value	NA



Arithmetic operators

Addition	+
Subtraction	-
Division	/
Power	$\wedge$
Scalar multiplication	*
Matrix multiplication	%*%



Syntax operators

Comment line	#
Assignment	<-
Access content	\$
Equal	=



Logical operators

Equal	==
Not equal	!=
Greater than	>
Greater than or equal to	>=
Less than	<
Less than or equal to	<=
contains	%in%
x AND y	x & y
x OR y	x y
NOT x	!x



Objects

Also called data structures

Vector	Only elements of the same class
List	Elements of any class
Factor	Categorical data
Matrix	Elements of the same class in 2D
Data frame	Elements of multiple classes in 2D
NULL	Empty object



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One dimension

Vector



List



Factor



↑
levels

↑
elements



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Vectors

- Can only contain objects of the **same class**
- Most basic type of R object
- Variables are vectors

`var1 ← 23`

→ 23 vector of length 1

`var2 ← "abc"`

→ abc vector of length 1



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Vectorized operations

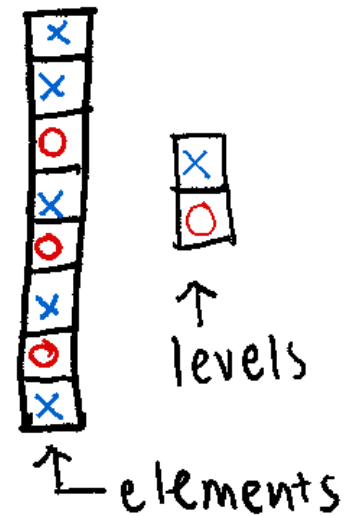
Diagram illustrating a vectorized addition operation. Two vertical vectors are added together to produce a result vector. The first vector contains the values [1, 2, 1, 4, 5, 7, 8, 2]. The second vector contains the values [2, 5, 3, 9, 7, 5, 6, 7]. The result vector contains the values [3, 7, 4, 13, 12, 12, 14, 9]. A blue arrow labeled $1+2$ points to the first element of the result vector, indicating the operation performed on the first elements of the input vectors. A blue dotted line with a plus sign connects the two input vectors.

Diagram illustrating a vectorized addition operation on 3x3 matrices. The first matrix contains the values $\begin{bmatrix} 3 & 6 & 8 \\ 2 & 1 & 5 \\ 7 & 4 & 9 \end{bmatrix}$. The second matrix contains the values $\begin{bmatrix} 5 & 1 & 7 \\ 3 & 4 & 6 \\ 8 & 9 & 2 \end{bmatrix}$. The result matrix contains the values $\begin{bmatrix} 8 & 7 & 15 \\ 5 & 5 & 11 \\ 15 & 13 & 11 \end{bmatrix}$. A blue arrow labeled $3+5$ points to the top-left element of the result matrix, indicating the operation performed on the top-left elements of the input matrices. A blue dotted line with a plus sign connects the two input matrices.



Factors

- Useful for categorical data
- Can have implicit order, if needed
- Each **element** has a label or **level**
- They are important in statistical modelling and plotting with ggplot
- Some operations behave differently on factors



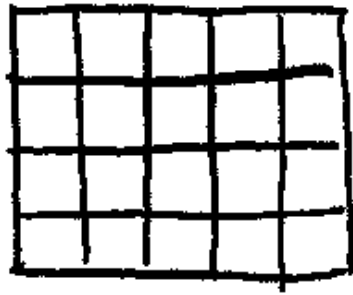
Lists

- Can contain objects of multiple classes
- Very important data type in R
- Extremely powerful when combined with some built-in functions



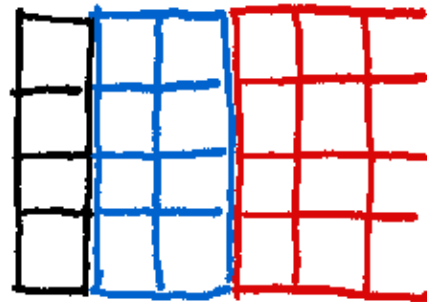
Multiple dimensions

Matrix



4x5

Data Frame



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Control structures and functions

Learning objectives:

- Understand the concept of environments in R
- Create new functions
- Implement conditional statements
- Implement a for loop to iterate over a list of files



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Conditional statements

- When we want a set of actions to be executed only if certain conditions are met

```
# if
if (condition is true) {
  perform action
}

# if ... else
if (condition is true) {
  perform action
} else { # that is, if the condition is false,
  perform alternative action
}
```



For loop

- Repeat a set of operations a certain number of times

```
for (iterator in set of values) {  
    do a thing  
}
```



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While loop

- Repeat a set of operations until a condition is no longer met

```
while(condition_is_true){  
    do a thing  
}
```



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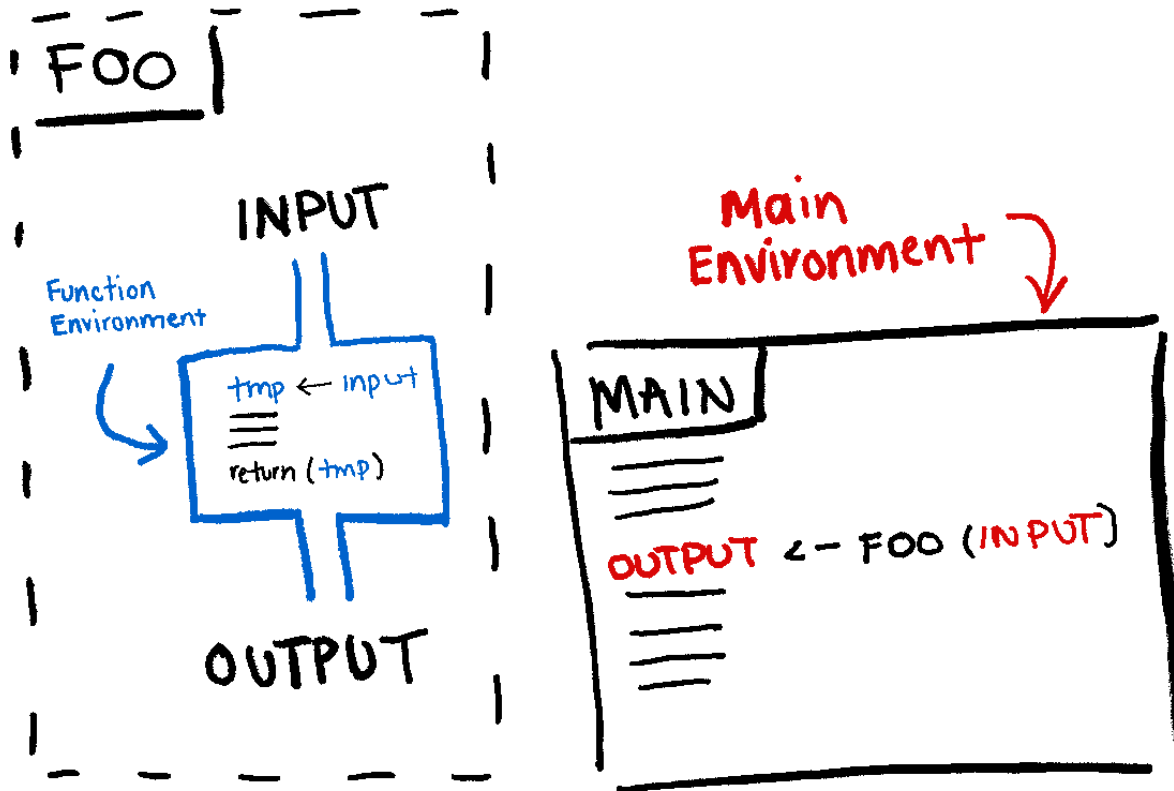
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What if base R is not enough?

- Sometimes your analysis requires tools that are not available in base R
- Two options:
 1. **Create new functions**
 2. **Packages** provide a way to incorporate methods and functions from



Functions and environments



Pass by value and scope

- When we pass an object to a function, a copy of it is created internally
- The changes made inside the function won't modify the original object we passed to it
- Any variables created inside the function will only exist during the function's execution time



Packages

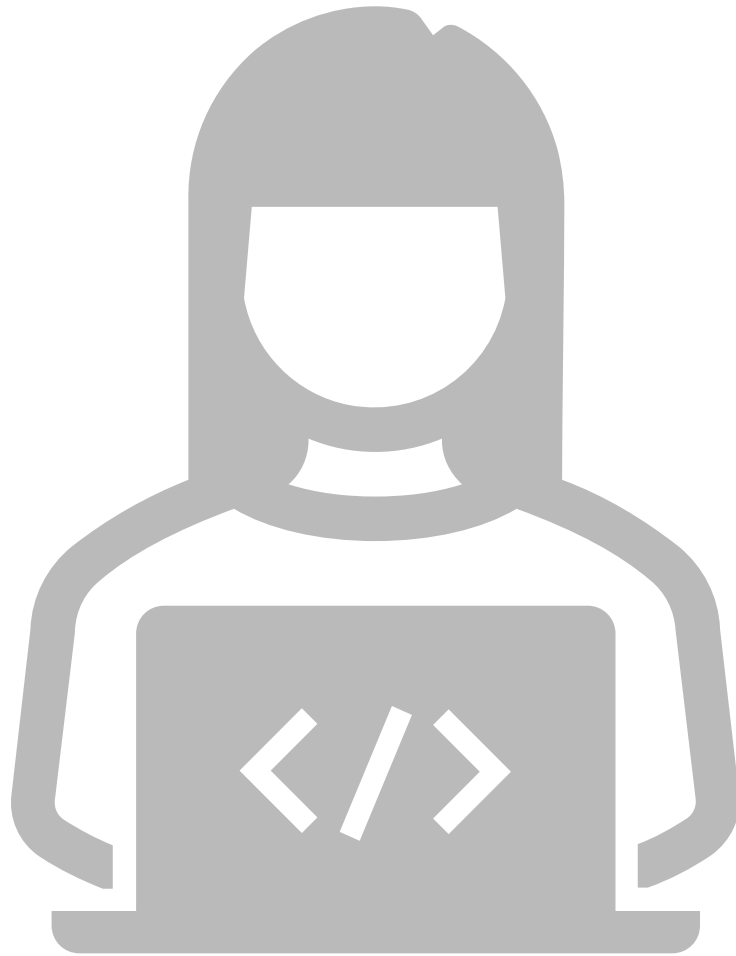
- Packages are a way for users to share methods they have developed
- Incorporate novel methods, datasets, or visualization tools
- Downloaded from many places:
 - **Comprehensive R Archive Network (CRAN)**
 - **Bioconductor**
 - GitHub, Bitbucket etc.



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Basic data manipulation

Learning objectives:

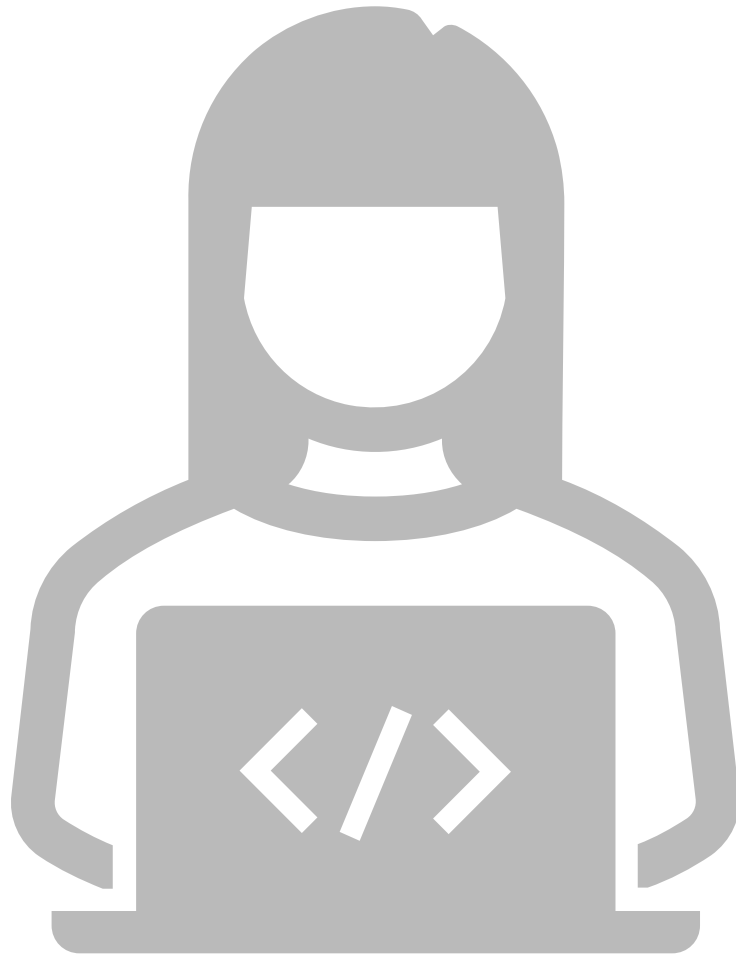
- Learn how to read/write data to/from files with different formats (.tsv, .csv)
- Familiarize with basic operations of data frames
- Index and subset data frames using base R functions
- Manipulate specific data frame columns
- Joining by columns and rows



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Advanced data manipulation

Learning objectives:

- Become familiar with the dplyr syntax
- Create pipes with the operator `%>%`
- Perform operations on data frames using dplyr and tidyr functions
- Implement functions from other external packages



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Split-Apply-Combine problem

INPUT

x	y
a	2
a	4
b	0
b	5

SPLIT

x	y
a	2
a	4

x	y
b	0
b	5

APPLY

x	y
a	3

x	y
b	2.5

COMBINE

x	y
a	3
b	2.5

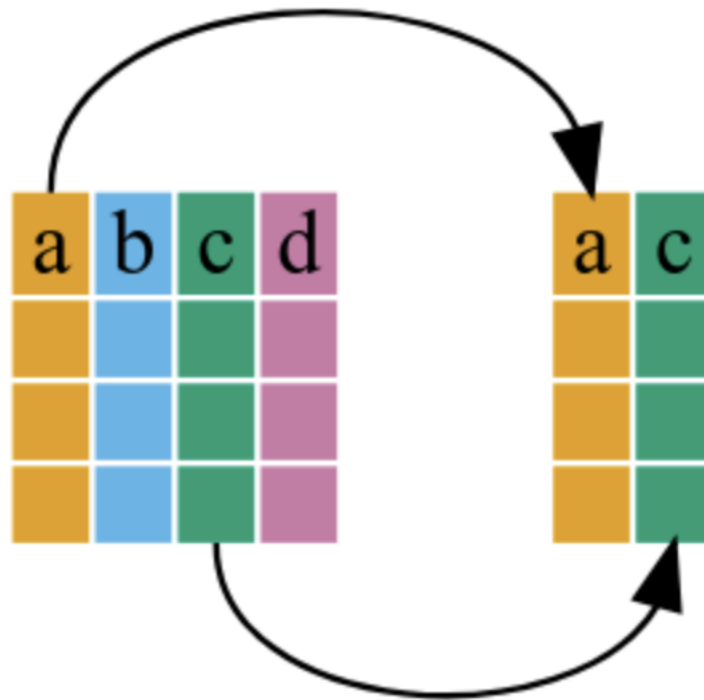


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Select

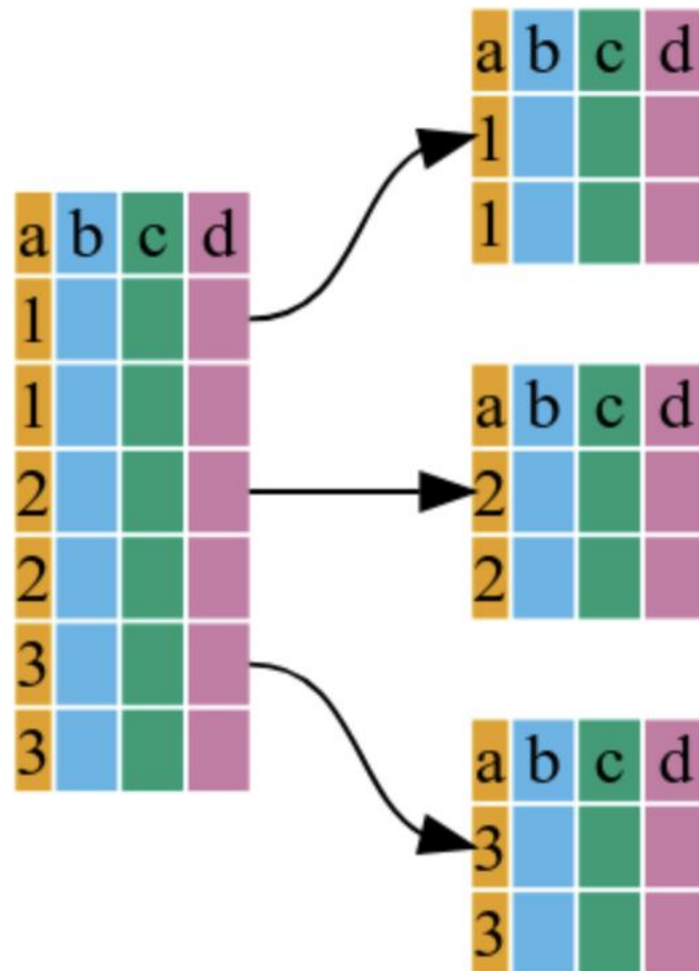


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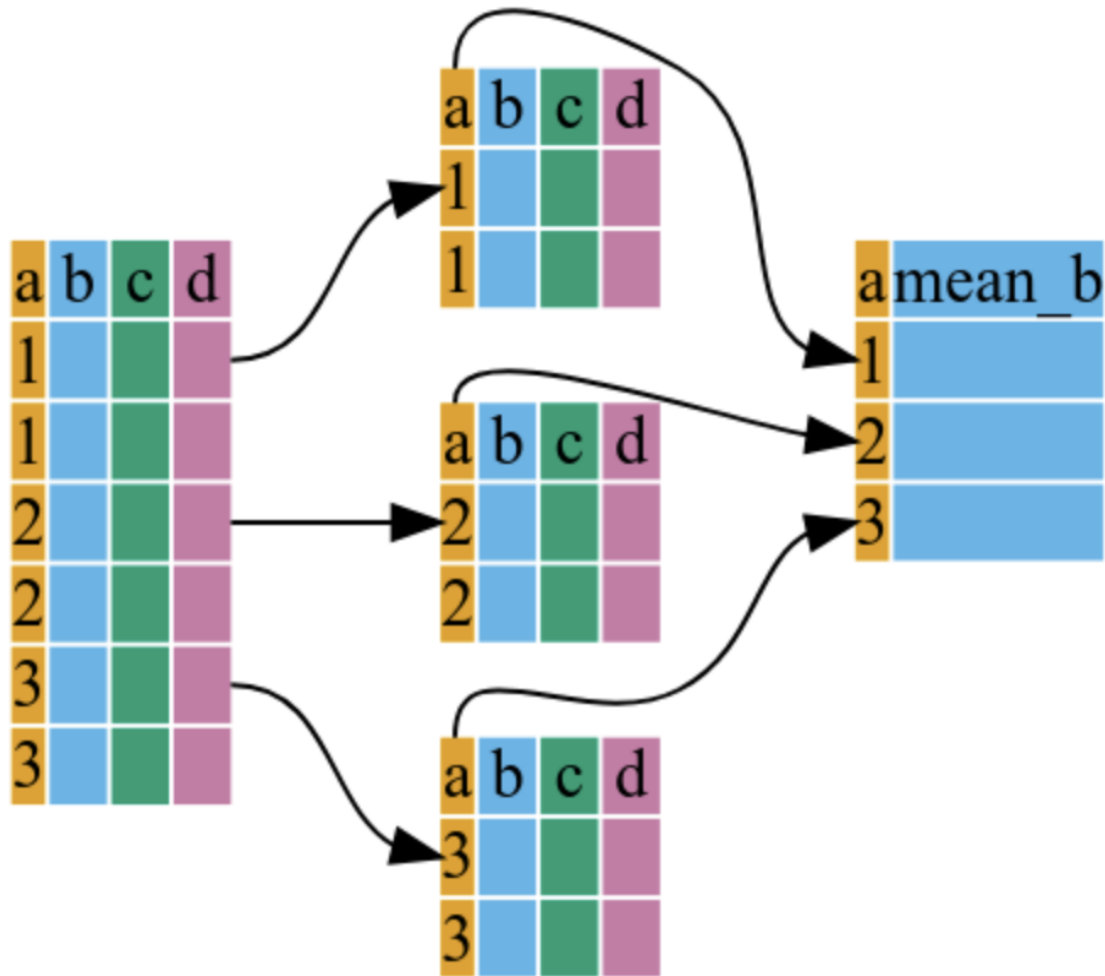
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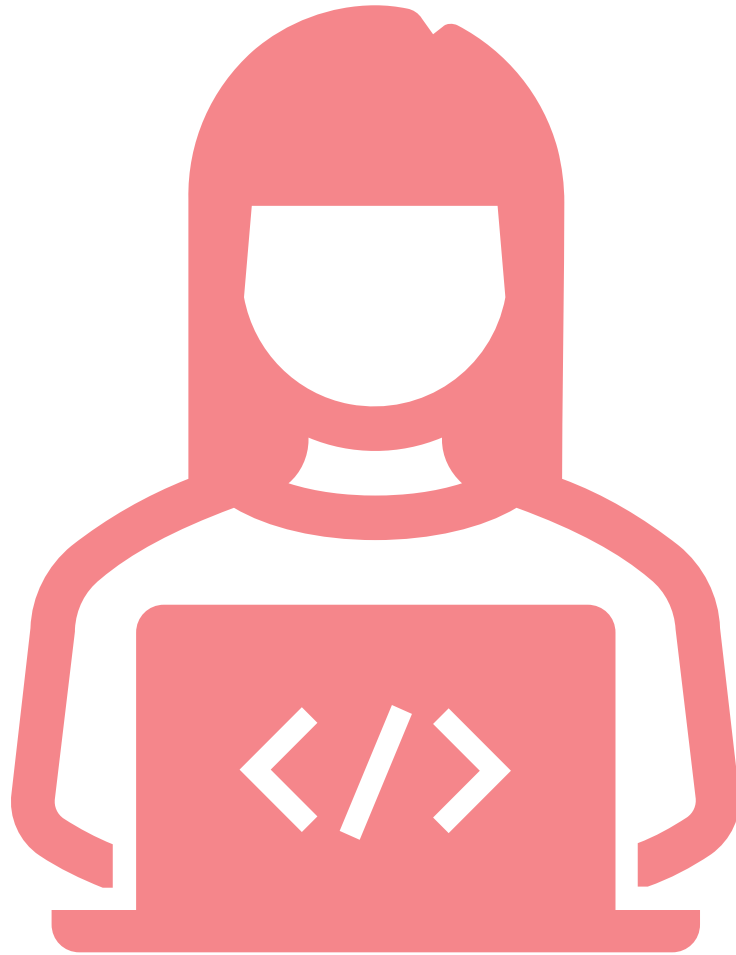
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Group by



Summarize





Break



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Generating visual outputs

Learning objectives:

- Create basic plots using base R functions
- Understand the connection between data frames and ggplot2
- Create basic graphs with ggplot2
- Use factors to customize graphics in ggplot2
- Learn about RMarkdown syntax to create reports
- Get familiar with existing RMarkdown templates



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Formatting data for ggplot

WIDE

	a	b	c
1	1	5	10
2	2	7	11
3	1	4	9

Values

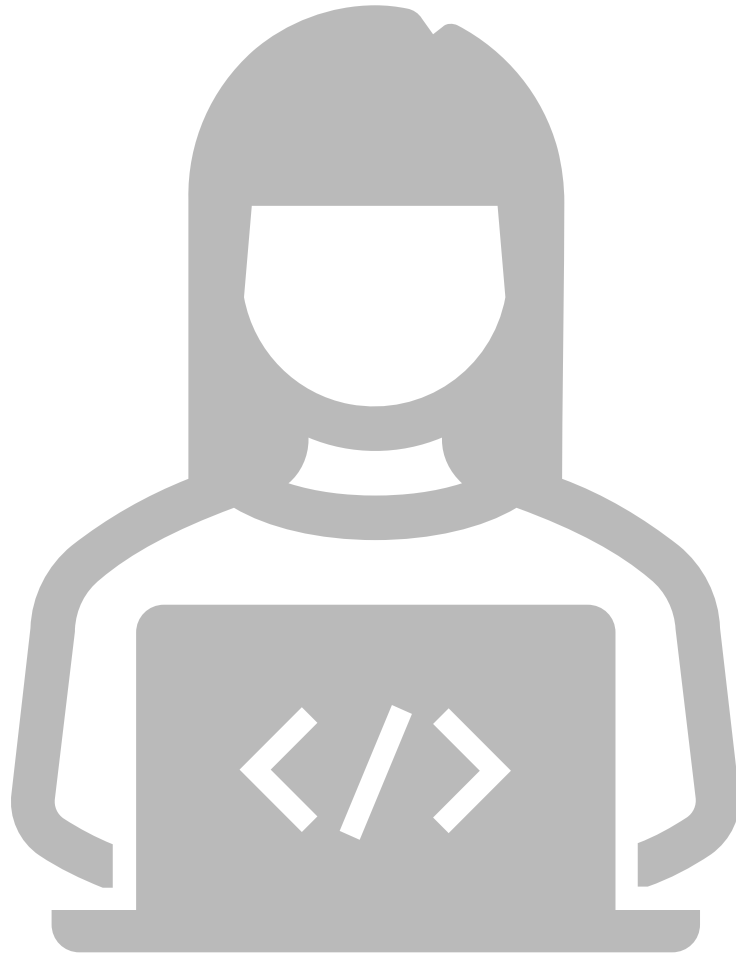
Variables

LONG

Values	variables
1	a
2	a
1	a
5	b
7	b
4	b
10	c
11	c
9	c

grouping factor





Activity: Analyzing a medical data set

Learning objectives:

- Familiarize with a real-life use case of R
- Apply the knowledge from previous modules to create an analysis pipeline



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COVID testing dataset

Details

Data on testing for SARS-CoV2 from days 4-107 of the COVID pandemic in **2020**. CHOP is a pediatric hospital in Philadelphia, Pennsylvania, USA. These data have been anonymized, time- shifted, and permuted.



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The dataset

Documentation

- Part of the medicaldata package
- https://htmlpreview.github.io/?https://github.com/higgi13425/medicaldata/blob/master/man/description_docs/covid_desc.html
- https://htmlpreview.github.io/?https://github.com/higgi13425/medicaldata/blob/master/man/codebooks/covid_testing_codebook.html



Format

A data frame with 15524 observations and 17 variables

subject_id id number for each subject; type: numeric

fake_first_name an auto-generated fake first name; type: character

fake_last_name an auto-generated fake last name; character

gender anonymized Gender, levels: female, male; type: character

pan_day day after start of pandemic; type: numeric

test_id test that was performed, levels: covid, xcvd1; type: character

clinic_name Clinic or ward where the specimen was collected, 88 levels; type: character

result result of test, levels: positive, negative, invalid; type: character

demo_group patient group, levels: patient, misc_adult, client, other adult, unidentified; type: character

age Age of subject at time of specimen collection (Anonymized), units = years; type: numeric

drive_thru_ind Whether the specimen was collected via a drive-thru site, levels: 1: Collected at drive-thru site; 0: Not collected at drive-thru site; type: numeric



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ct_result Cycle at which threshold reached during PCR, range: 14.05-45; type: numeric

orderset Whether an order set was used for test order, levels: 1: Collected via orderset; 0: Not collected via orderset; numeric

payor_group Payor associated with order, levels: commercial, government, unassigned, medical assistance, self pay, charity care, other; type: character

patient_class Disposition of subject at time of collection, levels: inpatient, emergency, observation, recurring outpatient, outpatient, not applicable, day surgery, admit after surgery-obs, admit after surgery-ip; type: character

col_rec_tat Time elapsed between collect time and receive time, range: 0 - 61370.2, units = hours; type: numeric

rec_ver_tat Time elapsed between receive time and verification time, range: -18.6 - 218.2, units = hours; type: numeric ...



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Software development concepts

Learning objectives:

- Familiarize with general good coding practices
- Learn about documentation standards
- Things to avoid when programming in R
- Learn how to debug and troubleshoot



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What we learned today:

- What is R
- Basic syntax, data types
- Data Manipulation and Visualization
- Package Installation



What's next?

Statistics in R Workshop (Part 2)

- Data Wrangling
- Regression
- Statistical Analysis

<https://www.mcgill.ca/micm/training/workshops-series>



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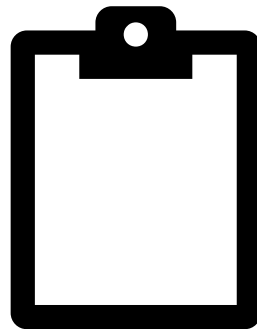
Thank you for attending!

1



Scan the QR code to confirm you attended today's workshop.

2



Fill out the feedback survey in the next 72h.

3



Get recognition for this workshop on your co-curricular record.



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Useful links

- [R software project](#)
- [RStudio Cheatsheet](#)
- [R ggplot2 Cheatsheet](#)
- [R dplyr Cheatsheet](#)
- [More resources](#)

